

FOOD WASTE INTELLIGENCE: AFRICA

From Food Loss to Food Rescue

Documentary Project Report

PROJECT	Food Waste Intelligence: Africa
AUTHOR	Francis Blessing Osewayeme
TOOLS	Power BI • SQL • Python • Excel
PRIMARY SOURCES	UNEP Food Waste Index • FAO Food Loss and Waste Database
RESCUE DATA	Synthetic / illustrative portfolio dataset
OUTPUT	Interactive 3-page Power BI dashboard + analytical workflow

Purpose: to demonstrate how data analytics can connect food-waste evidence, food-loss patterns and potential food-rescue operations into one decision-support story.

Independent portfolio project
September 2026

1. Executive Summary

Food Waste Intelligence: Africa is an end-to-end data analytics project examining three connected questions: where household food waste is being reported, where food loss occurs across the supply chain, and how surplus food could potentially be redirected through food-rescue operations.

The project combines study-level household food-waste observations from UNEP, food-loss observations from the FAO Food Loss and Waste Database, and a synthetic food-rescue dataset created for portfolio purposes. Descriptive evidence is separated from illustrative operational scenarios.

The central analytical idea is to move from measurement to action: Where is the problem? What patterns are visible? What intervention opportunities could be explored?

Project outputs

- A three-page Power BI dashboard covering Africa household food waste, the African food-supply chain and food rescue.
- Cleaned and structured datasets for UNEP, FAO and the synthetic rescue scenario.
- SQL analysis for data-quality checks, summaries and business questions.
- Python-based exploratory analysis and visualization workflow.

Scope at a glance

Component	Source / Type	Scale
UNEP household waste	UNEP 2024 study-level observations	49 extracted African observations
FAO food loss	FAO Food Loss and Waste Database	21,061 African observations
Food rescue	Synthetic / illustrative	100,000 operational-style records

2. Research Questions & Analytical Framework

Primary research question

Where are the major food-loss and household food-waste patterns being reported across Africa, and how could data help identify opportunities for food rescue and waste reduction?

Supporting questions

- Which African countries and study locations have the highest reported household food-waste observations?
- Which supply-chain stages and commodities have higher reported food-loss observations?
- What causes and activities are associated with reported food loss?
- Which cities, business types and food categories could represent food-rescue opportunities in the illustrative scenario?
- How can surplus food be translated into rescued meals, recovered revenue and estimated CO₂e impact?

Analytical framework

Layer	Analytical role	Main output
1. Household waste	Describe available UNEP evidence	Country/location comparisons
2. Food loss	Explore supply-chain patterns	Stage, commodity and cause analysis
3. Food rescue	Illustrate operational intervention	Rescue rate, meals, revenue and CO ₂ e

3. Data Sources & Data Governance

UNEP Food Waste Index

The household food-waste component uses observations from the UNEP Food Waste Index Report 2024. The extracted project table contains 49 African study-level observations.

Interpretation rule: a study-level observation is not automatically a nationally representative estimate. Country comparisons therefore use the wording “reported” and “available evidence.”

FAO Food Loss and Waste Database

The cleaned African FAO subset contains 21,061 observations. The data covers countries, commodities, years, activities, food-supply stages, treatments, causes, methods and references.

Critical analytical rule: loss percentages from separate supply-chain stages are not summed. The project instead uses average, median, maximum and observation-count measures where appropriate.

Synthetic Food-Rescue Dataset

The rescue component is intentionally synthetic. It contains fictional businesses and illustrative operational fields such as surplus kilograms, meals available, meals rescued, meals expired, rescue rate, recovered revenue and estimated CO₂e avoided.

Disclosure: this dataset is not operational data from Too Good To Go and the project does not claim affiliation, sponsorship or endorsement by UNEP, FAO or Too Good To Go.

Source links

Source	URL
UNEP Food Waste Index Report 2024	https://www.unep.org/resources/publication/food-waste-index-report-2024
FAO Food Loss and Waste Database	https://www.fao.org/platform-food-loss-waste/flw-data/en
FAO Food Loss Measurement	https://www.fao.org/platform-food-loss-waste/food-loss/food-loss-measurement/
Too Good To Go	https://www.toogoodtogo.com/

4. Data Preparation & Quality Controls

UNEP preparation

- Normalized country, region, study-area and source text fields.
- Converted annual food-waste values to numeric types.
- Recalculated monthly and daily equivalents from annual values.
- Added approximate latitude/longitude for study-area mapping; these are representative locations, not household GPS points.
- Flagged duplicate records for transparent review.

FAO preparation

- Normalized text fields and converted numeric fields to appropriate types.
- Created a unique record identifier.
- Created data-quality flags for loss percentage, loss quantity, sample size, source URL, activity, supply-chain stage and cause.
- Created global and African analytical subsets.
- Retained rows with missing values where deletion could introduce bias; missingness is exposed through quality flags.

Synthetic rescue preparation

- Generated 100,000 operational-style records across 10 African countries.
- Included city-level latitude and longitude for geographic visualization.
- Included multiple local currencies and an illustrative USD field for cross-country comparison.
- Marked every record as Synthetic so it cannot be confused with real company operations.

Data-quality principle

The project prioritizes traceability. Each major analytical component documents its source, scope, assumptions and limitations so that dashboard insights can be separated from the underlying evidence and from synthetic modelling assumptions.

5. Power BI Dashboard Documentation

Page 1 — Africa Food Waste (UNEP)

High-level view of reported household food-waste evidence across African study locations.

- Study observations, countries represented, average reported waste and highest reported waste.
- Country comparison and geographic distribution.
- Evidence-confidence and study-year views.

Page 2 — Africa Food Loss (FAO)

Examine reported food-loss patterns across the African food supply chain.

- Average reported loss by supply-chain stage.
- Top commodities and reported causes.
- Country comparisons and commodity-by-stage hotspot matrix.

Page 3 — From Food Waste to Food Rescue

Illustrate how operational data could identify surplus-food rescue opportunities.

- Surplus food, meals available, meals rescued and rescue rate.
- Illustrative recovered revenue and estimated CO₂e avoided.
- City, business-type, food-category and geographic views.

6. Illustrative Food-Rescue Scenario

The rescue dataset is designed to demonstrate what an operational analytics layer could look like if surplus-food rescue transactions were available. It is not presented as measured activity.

Metric	Illustrative value
Records	100,000
Countries	10
Estimated CO ₂ e avoided	1.92 million kg (illustrative dashboard snapshot)
Rescue rate	71.28% (illustrative dashboard snapshot)

Core measures

Measure	Definition
Total Surplus KG	Sum of surplus food weight.
Total Meals Available	Sum of meals potentially available for rescue.
Total Meals Rescued	Sum of meals successfully rescued in the synthetic scenario.
Rescue Rate	Meals rescued ÷ meals available.
Revenue Recovered USD	Illustrative recovered revenue normalized for cross-country comparison.
CO ₂ e Avoided	Illustrative estimated carbon-dioxide-equivalent impact avoided through rescue.

CO₂e means carbon dioxide equivalent. In this project, the measure is explicitly labelled as estimated and synthetic to avoid presenting it as a measured environmental outcome.

7. Key Findings & Interpretation Guidance

Household food waste

The UNEP component shows substantial variation among reported African study locations. The strongest portfolio interpretation is about the distribution and availability of evidence rather than a definitive ranking of countries as the continent's largest waste generators.

Dashboard wording: “Seychelles records the highest reported household food-waste observation in the extracted African study table at 183 kg/person/year.”

Food loss

The FAO component focuses on supply-chain stages, commodities, activities and causes represented in the reported observations. Loss percentages are not added across stages.

Food rescue

The synthetic rescue layer illustrates how operational analytics could connect surplus food to rescue rates, recovered revenue and environmental impact, bridging descriptive measurement and potential intervention.

8. Portfolio & GitHub Documentation Checklist

Asset	Recommended location
Power BI .pbix	/powerbi/
Dashboard screenshots	/screenshots/
UNEP cleaned CSV	/data/unesco/
FAO Africa cleaned CSV	/data/fao/
Synthetic rescue CSV	/data/synthetic/
Python notebook	/python/
SQL scripts	/sql/
Data dictionary	/documentation/
Project report	/documentation/

Conclusion

Food Waste Intelligence: Africa demonstrates a complete analytics workflow: defining a real-world problem, sourcing data, cleaning and modelling it, building interactive Power BI dashboards, and documenting assumptions. Its central contribution is the connection between data → insight → intervention → impact.

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